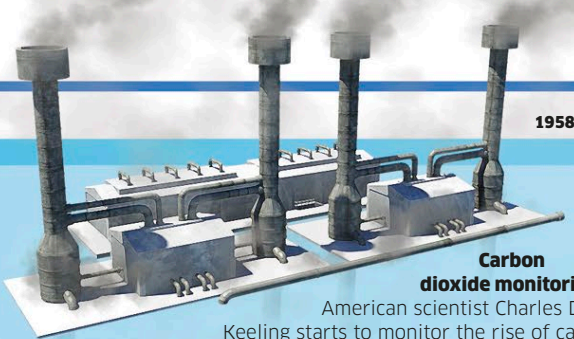


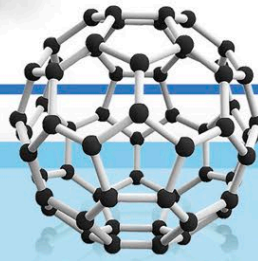
**1661** Robert Boyle's *The Sceptical Chymist* develops a theory of atoms.



1958

### Carbon dioxide monitoring

American scientist Charles David Keeling starts to monitor the rise of carbon dioxide in the atmosphere. His Keeling Curve graph is still used to study climate change.



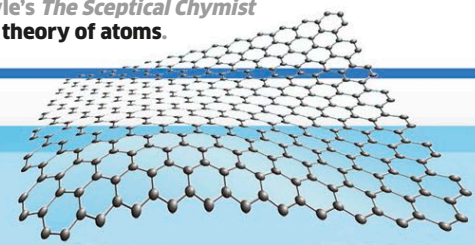
1985

### BUCKYBALL

### Buckyball discovery

Scientists at Rice University in Houston discover a new form of carbon called buckminsterfullerene, or buckyball.

2004



### World's thinnest material

Graphene (a layer of carbon atoms just one atom thick) is produced at the University of Manchester, UK. It is the world's thinnest material, but 200 times stronger than steel.

### GRAPHENE

**1945 - PRESENT**

1870

### Synthetic materials

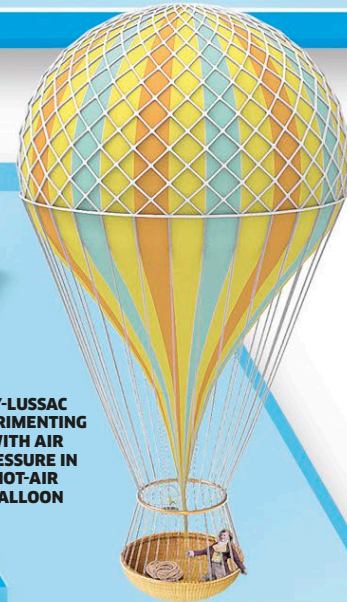
The first synthetic materials made from cellulose are invented: celluloid (moldable plastic) in 1870 and viscose rayon in 1890.



1869

### Mendeleev's periodic table

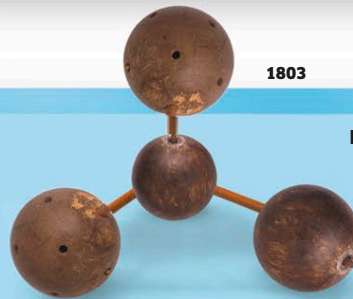
Russian chemist Dmitri Mendeleev arranges the 59 known elements into groups based on their atomic mass and properties. This periodic table enables him to predict the discovery of three more elements.



### GAY-LUSSAC EXPERIMENTING WITH AIR PRESSURE IN HOT-AIR BALLOON

### VISCOSE, SYNTHETIC SILK

1803



### DALTON'S ATOMIC MODELS

### Dalton's atomic theory

English chemist John Dalton argues that all matter is composed of atoms and atoms of the same element are identical. He compiles a list of elements based on their atomic mass, then known as atomic weight.

### Structure of water

French chemist Joseph Louis Gay-Lussac experiments with gases and pressure and finds that water is made up of two parts hydrogen and one part oxygen.

1805

### Industrial Revolution

Driven by the thirst for modernization, chemists identified more elements and invented ways to use them in medicine, in creating new materials, and in advanced industrial technologies.

**1800 - 1890**

1527

### Salts, sulfurs, and mercuries

Swiss chemist Theophrastus von Hohenheim works out a new classification for chemicals, based on salts, sulfurs, and mercuries.



### Age of Discovery

The Renaissance brought both rediscovery of antique knowledge and a quest for fresh ideas. Scientists began to test, experiment, and document their ideas, publishing their findings and working hard to classify matter.

**17TH CENTURY**

**1600 - 1800**

### Middle Ages

In Asia and the Islamic world, alchemists experimented to find the elixir of life and to make gold. By the late Middle Ages, European alchemists were working toward the same goal.



### Gunpowder

While they are looking for the elixir of life, Chinese alchemists accidentally invent gunpowder by mixing saltpeter with sulfur and charcoal.

### Classifying elements

Arab physician Al-Razi divides elements into spirits, metals, and minerals depending on how they react with heat.



**500 CE - 1600**

855 CE

900